

Notifications & Reminders

Kharka Farm OS · Livestock monorepo · Prepared 3 Oct 2026
Covers the API (`apps/api`), mobile app (`apps/mobile`) and web Inbox (`apps/web`).

In one paragraph. Every reminder is a *task* in the farm's Inbox. Tasks are created automatically from farm events (a calving, a heat, a service, low stock, an unpaid sale) or by scheduled checks. A dispatcher then delivers each due task as a **push notification**, and for the most urgent ones also by **SMS** and a recorded **Nepali voice call**. Anti-fatigue rules hold back non-urgent alerts at night, cap how many a farmer gets per day, and bundle similar ones together. Urgent alerts that nobody acknowledges escalate to the manager, then the owner. The phone also keeps alarms for urgent tasks so they still ring with no signal.

1. How a reminder reaches the farmer

- A task is created.** Triggered by an event (e.g. recording a calving creates the colostrum reminders), or by the nightly / hourly checks (e.g. stock below reorder level). Creating the same task twice changes nothing.
- It appears in the Inbox** on web and mobile, with a button that opens the right form.
- The dispatcher sends it** when it falls due, applying the delivery rules in section 2.
- The phone schedules an offline alarm** for urgent task types, so it fires even without internet.
- The farmer acts** — Do, Done, Snooze or Dismiss — and the task closes.

Delivery channels

Channel	Used for	How it works
Push notification	Every reminder	Sent through Firebase Cloud Messaging (installed app) or the Expo push service (Expo Go testing). The phone registers its push token after login. Android channels: <i>Critical farm alerts</i> (max importance) and <i>Farm reminders</i> .
SMS	Critical alerts, and High alerts overdue by 24 h	Sent through Sparrow SMS (Nepal, NTC + Ncell), with Twilio as a fallback. Goes to the mobile number each person saves in the Inbox <i>SMS alerts</i> card, or that an admin sets on the Members page (Nepal mobiles only, e.g. 98XXXXXXX). Capped at 40 SMS per farm per Nepali month (<code>SMS_MONTHLY_CAP</code>). Nepali text uses 70 characters per SMS segment, English 160.
Voice call	Critical alerts and escalations	Built from pre-recorded Nepali voice clips spliced together (e.g. "colostrum" + "number is" + digits). Never machine text-to-speech. Queued for now; not dialled yet.
Offline phone alarm	Urgent types (section 5)	Scheduled on the phone itself from tasks synced in the last 30 days, so it rings with no signal.
Inbox	Everything	Web and mobile list of open tasks with actions. Works offline from the last synced copy.

Who gets which reminder

Reminder	Sent to	Examples
CRITICAL (any type)	Everyone: admin, manager and all workers	Calving today, colostrum, retained placenta, urgent vet, calf health check.

Shed work	Workers only. If the farm has no workers, the manager; if no manager either, the admin.	Heat watch, service window, vaccination, medication dose, dry-off, milk withhold end, missing milk log, markers, sync injections.
Decisions & money	Admin and manager	Expense approval, unpaid revenue, stock reorder, expiring lots, stock count, tank variance, yield drop, repeat breeder, anestrus vet / decision, protocol broken, cycling unbred, pregnancy check stalled.
Assigned task	Only the person it's assigned to (Critical still goes to everyone)	A vaccination assigned to one worker.
Farm vet	Only animals that need a vet	Urgent vet, anestrus vet, treatment follow-up, pregnancy check, calf health check.

2. Delivery rules (anti-fatigue)

Rule	What it does
Quiet hours 20:00 – 05:00	Non-critical alerts are held and delivered at 05:00 Nepal time. CRITICAL alerts always go through immediately.
Buffalo silent-heat exception	The buffalo silent-heat check is allowed out at 04:00 , because buffalo show heat mostly before dawn. On by default for buffalo farms; can be switched off per user.
Daily cap	At most 5 non-critical pushes per person per day . Extra ones stay in the Inbox but don't buzz the phone.
Grouping	Same-type alerts are bundled into one push, e.g. "6 animals to check for heat this morning" or "4 animals due for FMD". Never grouped: colostrum, calving watch, urgent vet, and sync-protocol injections — one animal, one alert.
SMS gate	SMS only for Critical alerts, or High alerts more than 24 h overdue, and only while the monthly cap has room.
Escalation	An unacknowledged Critical alert escalates to the manager after 30 minutes and the owner after 60 minutes .
Mute offer	After a user dismisses the same reminder type 3 times , they're offered to mute it for 14 days . Never offered for Critical types.
Expiry	Open tasks more than 30 days past due are marked expired by the nightly job.

When things run

Job	Schedule (Nepal time)	Purpose
Dispatcher	Every 5 minutes	Sends due tasks as push / SMS / voice, applying the rules above.
Hourly job	Every hour, on the hour	Service windows closing / missed; escalates unacknowledged Critical alerts.
Nightly job	01:00 daily	Breeding stages and stage reminders, anestrus ladder, protocol checks, vaccinations, withhold ends, stock, missing milk, finance, profit metrics, and expiring stale tasks.

On Vercel: serverless hosting doesn't run timers. The dispatcher is triggered every 5 minutes by the GitHub Action `notification-dispatch.yml` calling `POST /v1/notifications/dispatch`. The **hourly and nightly jobs do not yet have an external trigger on Vercel**, so in production their reminders are only created by in-app events until one is added.

3. Reminder catalogue

CRITICAL push + SMS (+ voice where noted), escalates · **HIGH** push; SMS if 24 h overdue · **NORMAL** push · **LOW** push, lowest priority · **NEW** added in the latest release.
 [tag] is the animal's short herd number.

3.1 Calving & newborn care

Reminder	When	Urgency	Channels	Message (English)	Message (Nepali)
Calving in 7 days	7 days before expected calving, 08:00	HIGH	Push	[tag] calving in 7 days	[tag] ७ दिनमा बियाउने
Calving in 3 days	3 days before, 08:00	CRITICAL	Push, SMS	[tag] calving in 3 days	[tag] ३ दिनमा बियाउने
Calving tomorrow	1 day before, 08:00	CRITICAL	Push, SMS	[tag] calving tomorrow	[tag] भोलि बियाउने
Calving today	Due date, 06:00	CRITICAL	Push, SMS	[tag] due to calve today	[tag] आज बियाउने
Calving overdue	3 days past due date	HIGH	Push, SMS	[tag] is 3 days past the due date	[tag] तोकिएको मितिभन्दा ३ दिन ढिलो
Calving well overdue	Well past due date	CRITICAL	Push, SMS, Voice	[tag] is well past the due date — check for a lost pregnancy or an unrecorded calving	[tag] धेरै ढिलो — गर्भ गएको वा बियाएको तर नलेखिएको हुन सक्छ
Colostrum 1	2 h after calving	CRITICAL	Push, SMS, Voice	Colostrum for [tag] — feed within 6 hours of birth	[tag] लाई बिगौती — जन्मको ६ घण्टाभित्र खुवाउनुहोस्
Colostrum 2	6 h after the first feed	CRITICAL	Push, SMS	Second colostrum for [tag]	[tag] को दोस्रो बिगौती
Colostrum 3	8 h after the second feed	HIGH	Push	Third colostrum for [tag]	[tag] को तेस्रो बिगौती
Retained placenta	12 h after calving	CRITICAL	Push, SMS, Voice	Check [tag] — placenta should have expelled by now	[tag] जाँच — खेर अहिलेसम्म झरिसक्नुपर्छ
Post-calving check	Day 7 and day 21, 08:00	NORMAL	Push	Day-7 / Day-21 post-calving check [tag]	[tag] बियाएको ७ / २१ दिन जाँच
High-risk calf	Daily from day 1 if colostrum was late, up to 21 days	HIGH	Push	High-risk calf [tag] — daily check	जोखिममा बाच्छा [tag] — दैनिक जाँच

3.2 Heat & breeding

Reminder	When	Urgency	Channels	Message (English)	Message (Nepali)
Wait period ending	3 days before the post-calving wait ends, 08:00	LOW	Push	[tag] — voluntary wait ends in 3 days. Start watching for heat.	[tag] — पर्खाइ ३ दिनमा सकिन्छ। गर्मी हेर्न सुरु गर्नुहोस्।

Daily heat watch	Every morning 05:00 while awaiting heat	NORMAL	Push	Heat watch [tag] this morning	आज बिहान [tag] को गर्मी हेर्नुहोस्
Next heat expected	2 days before the next expected heat, 05:00	HIGH	Push	[tag] should be in heat again in 2 days	[tag] फेरि २ दिनमा गर्मीमा हुनुपर्छ
Service window open	When a heat is recorded	HIGH	Push, SMS	Breed [tag] now — the service window is open	[tag] लाई अहिले सेवा दिनुहोस् — समय खुल्यो
Service window closing	3 h before the window closes	HIGH	Push, SMS	Service window for [tag] closes in 3 hours	[tag] को सेवा समय ३ घण्टामा बन्द हुन्छ
Service window missed	After the window closes with no service	NORMAL	Push	Service window missed for [tag] — watch for the next heat	[tag] को सेवा समय सकियो — अर्को गर्मी हेर्नुहोस्
Return-to-heat watch	After a service, around the next cycle, 05:00	NORMAL	Push	Heat watch [tag] — if she cycles, the service failed	[tag] गर्मी हेर्नुहोस् — आएमा सेवा असफल
Cycling but unbred	3rd heat this lactation with no service	HIGH	Push	[tag] — three heats and no service	[tag] — तीन पटक रजस्वला, सेवा छैन
Repeat breeder	On the 3rd service	HIGH	Push	[tag] is on service [n]. Check: energy deficit, uterine infection, AI timing, semen handling. Cumulative AI cost: NPR [cost].	[tag] को [n] औं सेवा। दानामा उर्जा, संक्रमण, एआई समय, वीर्य जाँचुहोस्।
Repeat breeder — vet	14 days after the repeat-breeder flag, still open	HIGH	Push	[tag] is still open 14 days after the repeat-breeder flag. Call the vet.	[tag] १४ दिनपछि पनि खुला छ। पशु चिकित्सक बोलाउनुहोस्।

3.3 Pregnancy & dry-off

Reminder	When	Urgency	Channels	Message (English)	Message (Nepali)
Pregnancy check due	Species pregnancy-check day after service, 08:00	HIGH	Push	Pregnancy check [tag]	[tag] को गर्भ जाँच
Pregnancy check overdue	14 days after it was due	HIGH	Push, SMS	Pregnancy check overdue — [tag]	गर्भ जाँच ढिलो — [tag]
Never checked	45 days after it was due	HIGH	Push, SMS	Bred [n] days ago and never checked. She may be carrying — do not breed again until examined.	[n] दिनअघि सेवा, जाँच भएको छैन। जाँच नगरी फेरि सेवा नदिनुहोस्।
Dry-off due	Species dry-off date before calving, 08:00	HIGH	Push, SMS	Dry off [tag] — confirm she is still pregnant first	[tag] को दूध बन्द गर्नुहोस् — पहिले गर्भ पक्का गर्नुहोस्
Dry-off overdue	Past dry-off date and still milking	HIGH	Push, SMS	[tag] is still milking — dry-off is overdue	[tag] अझै दूध दिँदैछ — दूध बन्द गर्न ढिलो भयो

Transition feed	21 days before calving, 08:00	NORMAL	Push	Transition feed for [tag] — 21 days to calving	[tag] को दाना बदल्नुहोस् — बियाउन २१ दिन
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3.4 No heat (anestrus) ladder

Reminder	When	Urgency	Channels	Message (English)	Message (Nepali)
Keep watching	When no heat is suspected	NORMAL	Push	[tag] has not shown heat — keep watching	[tag] लाई गर्मी आएको छैन — हेर्दै रहनुहोस्
Silent heat — buffalo	Every 3 days at 04:00	NORMAL	Push	Check [tag] between 4 and 7am — buffalo silent heat	[tag] बिहान ४ देखि ७ बजेसम्म हेर्नुहोस् — मौन गर्मी
Silent heat — cow	Every 3 days at 05:00	NORMAL	Push	Check [tag] at dawn — possible silent heat	[tag] बिहानै हेर्नुहोस् — मौन गर्मी हुन सक्छ
Try minerals	30 days into no-heat	NORMAL	Push	[tag] has not shown heat for [days] days. Try a daily mineral mixture for 3 weeks before calling the vet — about NPR 400.	[tag] लाई [days] दिन गर्मी छैन। ३ हप्ता खनिज मिश्रण दिनुहोस् — करिब रु ४००।
Call the vet	60 days into no-heat	HIGH	Push, SMS	[tag] — [days] days without heat. Call the vet.	[tag] — [days] दिन गर्मी छैन। पशु चिकित्सक बोलाउनुहोस्।
Decision point	90 days into no-heat	HIGH	Push, SMS	[days] days without heat. Delay is costing about NPR [cost]. Options: CIDR protocol, continue waiting, or sell.	ढिलाइ करिब रु [cost]। विकल्प: सिडर, पर्खिने, वा बेच्ने।

3.5 Synchronisation protocols & heifers

Reminder	When	Urgency	Channels	Message (English)	Message (Nepali)
Protocol injection	On each protocol step day	HIGH	Push, SMS	Injection due for [tag]	[tag] को सुई दिनुहोस्
Injection late	6 h after a missed step	CRITICAL	Push, SMS, Voice	Injection for [tag] is 6 hours late — the protocol will break if this is missed	[tag] को सुई ६ घण्टा ढिलो — छुट्यो भने प्रोटोकल रद्द हुन्छ
Protocol AI day	On the protocol's AI day	CRITICAL	Push, SMS	AI due today for [tag]	[tag] लाई आज एआई दिनुहोस्
Protocol broken	Nightly, when a step was missed	HIGH	Push	Protocol for [tag] is void — a missed injection. Restart, or switch to CIDR.	[tag] को प्रोटोकल रद्द — सुई छुट्यो।
Heifer ready	When a heifer reaches breeding age	NORMAL	Push	[tag] is old enough to breed	[tag] प्रजनन उमेर पुग्यो

Weigh heifer	Every 30 days while ready	NORMAL	Push	Weigh [tag] — heifers need a target weight before first service	[tag] तौलनुहोस् — पहिलो सेवाअघि तौल पुग्नुपर्छ
Heifer overdue	180 days after ready, still unbred	HIGH	Push	[tag] has been ready to breed for 180 days. Accumulated rearing cost: NPR [cost].	[tag] १८० दिनदेखि प्रजनन तयार छ।

3.6 Health & treatment

Reminder	When	Urgency	Notes
Vaccination due	Nightly, from the vaccination schedule	HIGH	Grouped by disease, e.g. "4 animals due for FMD". Offline phone alarm.
Medication dose	Each scheduled dose of a treatment	HIGH	Offline phone alarm.
Treatment follow-up	Follow-up date set on a treatment	HIGH	Opens the treatment form.
Urgent vet	Retained placenta and other emergencies	CRITICAL	Push, SMS, voice; never grouped; escalates.
Milk withhold ends	When an antibiotic milk withhold expires	NORMAL	"Withhold ends for [tag]" — milk can go back in the tank.

3.7 Milk, shed & stock

Reminder	When	Urgency	Notes
Missing milk record	Nightly, for milking animals with no milk entry after 20:00	LOW	"No milk recorded for [tag]".
Yield drop	When an animal's yield falls sharply	HIGH	Opens the Shed.
Tank variance	When the tank reading doesn't match recorded milk	HIGH	Opens the Shed.
Apply / remove marker	Marker cohort changes	HIGH	Can only be completed by scanning the tag.
Re-tag required	Lost or unreadable tag	HIGH	Opens Animals.
Low stock	Nightly, item below its reorder level	NORMAL	"Low stock: [item]". Offline phone alarm.
Lot expiring	Nightly, item or lot near expiry	HIGH ≤ 7 days, else NORMAL	"[item] lot [n] expires [date]".
Stock reconcile	Nightly, a lot shows negative quantity	HIGH	"Lot is negative — record the purchase".

3.8 Finance **NEW**

Reminder	When	Urgency	Notes
Expense approval	Nightly, for each expense pending or escalated	HIGH	"Expense approval: [category] ([amount])". Opens Expenses. Offline phone alarm.
Unpaid revenue	Nightly, for each sale with payment pending or partial	NORMAL	"Unpaid revenue: [source] ([amount])". Opens Revenue. Offline phone alarm.

Heat stress (temperature-humidity index above 78) is defined as a High alert but has no weather data source yet, so it doesn't fire.

4. Inbox actions

- **Do** — opens the right form directly (e.g. colostrum form, pregnancy check, Expenses).
- **Done** — closes the task. Works offline; it syncs when back online.
- **Snooze** — +1 hour, +1 day or +1 week.
- **Dismiss** — with a reason: not needed, already done offline, animal sold, wrong animal, other. Works offline.
- **Active / Completed tabs** and a **Mine only** filter.
- **Marker tasks** can only be closed by scanning the animal's tag.
- **Mute offer** appears after 3 dismissals of the same type.
- Titles show in the user's chosen language (English or Nepali).

Where "Do" takes you

Opens	Reminder types
Breeding forms	Colostrum, calving watch, heat watch, silent heat, service window, pregnancy check, repeat breeder, protocol injections and AI, anestrus steps, never-checked pregnancy
Health	Vaccination, medication dose, treatment follow-up, urgent vet, post-calving and calf checks
Shed	Missing milk, yield drop, tank variance, milk withhold end, markers
Inventory	Low stock, lot expiring, stock reconcile
Animals / Feed	Dry-off, re-tag / transition feed
Expenses / Revenue	Expense approval / unpaid revenue

5. Offline phone alarms

The app schedules alarms on the phone for these types, using tasks synced in the last 30 days. They ring at the due time even with no internet:

Colostrum · Calving watch · Urgent vet · Vaccination due · Medication dose · Service window · Low stock · Expense approval · Unpaid revenue

Heat-watch reminders are deliberately not offline alarms because they come every morning for many animals; they arrive as push notifications when online.

6. Production setup checklist

Setting	Where	Needed for
<code>FCM_SERVICE_ACCOUNT_JSON</code> (one-line Firebase key)	Vercel env	Push to installed (EAS) builds. Expo Go push works without it.
<code>NOTIFICATION_CRON_SECRET</code>	Vercel env + GitHub secret	Lets the GitHub Action trigger the dispatcher.
<code>API_BASE_URL</code> = <code>https://evoqedlivestockfarm.vercel.app</code>	GitHub secret	Dispatcher trigger target.
<code>ENABLE_NOTIFICATION_CRON</code> = <code>true</code>	GitHub variable	Turns the 5-minute dispatcher trigger on.
<code>SPARROW_SMS_TOKEN</code> , <code>SPARROW_SMS_FROM</code>	Vercel env	SMS through Sparrow SMS (recommended for Nepal). Used first when set.

<code>TWILIO_ACCOUNT_SID</code> , <code>TWILIO_AUTH_TOKEN</code> , <code>TWILIO_FROM_NUMBER</code>	Vercel env	Fallback SMS provider (optional; SMS is only logged if neither is set).
Each person's mobile number	Inbox → SMS alerts, or Members page	Without a saved number, that person gets push only.
<code>SMS_MONTHLY_CAP</code>	Vercel env	Optional; defaults to 40 per farm per month.
External trigger for hourly + nightly jobs	To build	Scheduled reminders (stock, vaccinations, breeding ladder, finance) on Vercel.

Source of truth: `apps/api/src/notifications/notification-rules.ts` (delivery rules and urgency), `apps/api/src/breeding/reminder-catalog.ts` (breeding and calving reminders), `apps/api/src/jobs/nightly.job.ts` and `hourly.job.ts` (scheduled checks), `apps/mobile/src/core/notify.ts` (offline phone alarms). Farms can override breeding reminder rules per farm; this document lists the system defaults.